Event Pattern-Instance Graph

A Multi-Round Role Representation Learning Strategy for Document-Level Event Argument Extraction

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Findings of ACL 2025

Presented by Aryan Damaraju

What Is the Task?

- Document-level event argument extraction
- Given an event trigger, find the text that fills each role
  - For an attack: attacker, instrument, target, place

> "…Tomahawk cruise missiles launched from a British **submarine** stationed in the Mediterranean Sea **struck** an administrative **building** in [Khadafy's] **Bab al-Azizia compound**."

Why it's hard

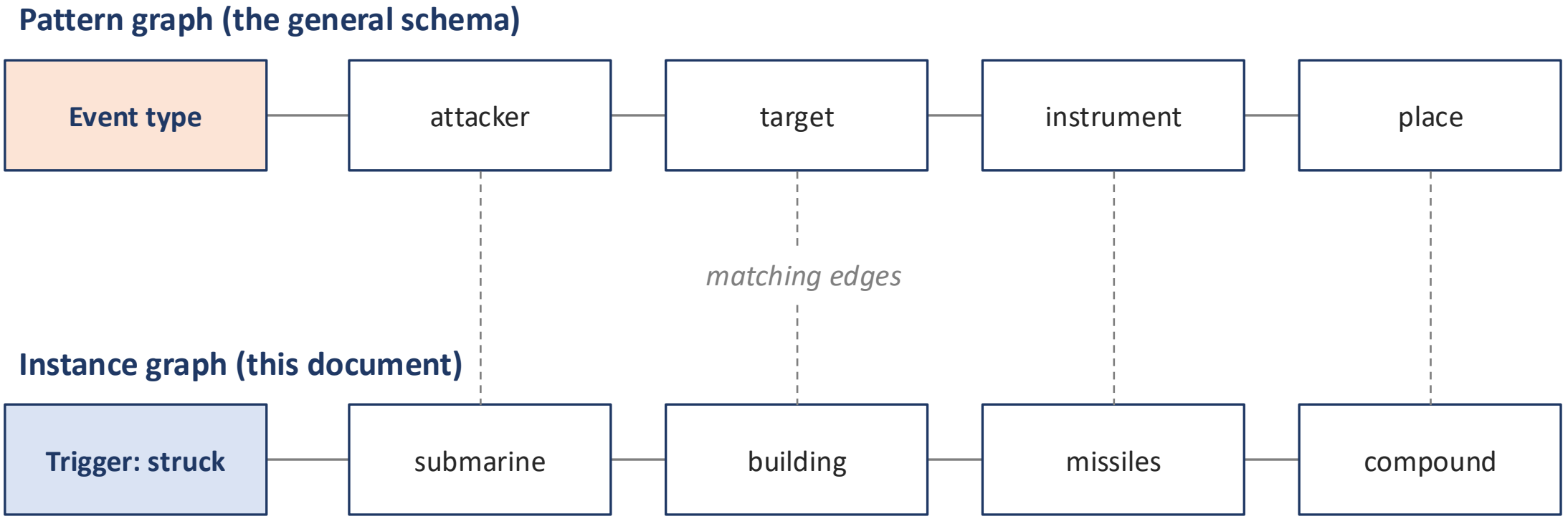
- Details can be far from the trigger
- Roles depend on each other
- Must link the general event type to this document

The Problem With Earlier Models

- Models like PAIE, TabEAE, DEEIA, and HMPEAE prompt a language model to get a vector for each role, then pick a span
- Problem 1: each role is predicted on its own, so how roles relate is ignored
- Problem 2: the general event type and the specific document are never linked directly

The Idea: EPIG

Two graphs, linked together



How the Model Works

1. BART reads the document and a prompt for each role
2. Make a rough first guess for each span
3. Put the guesses into the graph
4. A graph attention network updates the nodes and edges
5. Mix in a memory of each role from past documents
6. Predict the final spans

Updating the Graph

- Nodes: each node weighs its neighbors with attention, then updates
- Edges: each edge is averaged with the role, trigger, and span it connects
  - Older graph models used one fixed matrix per edge type. Here, edges carry context.
- Repeat for several layers
- Combine node and edge into the final role vector

Memory and Multiple Rounds

Role memory

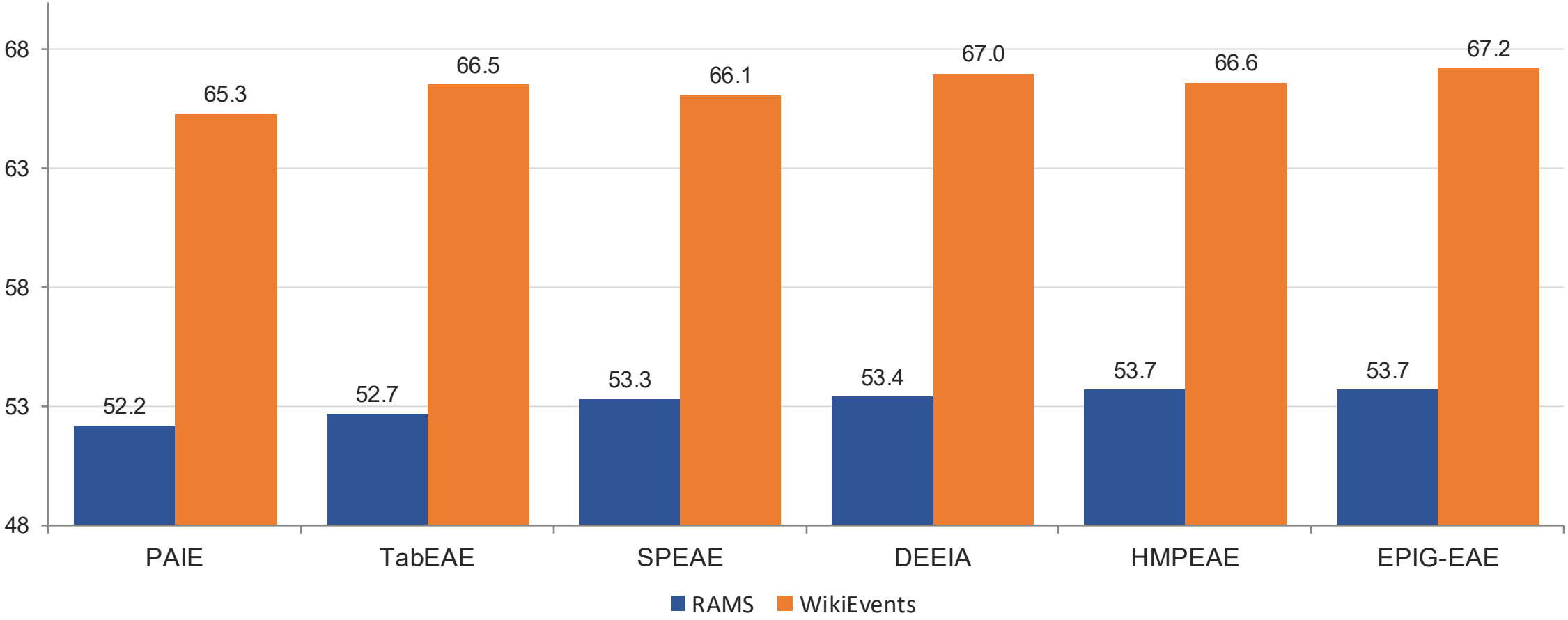
- One slot per role, updated on every training document
- Helps with roles the model rarely sees

Multiple rounds

- Round 1 spans go into the graph, which gives better role vectors
- Then predict the spans again
- Best at round 2. More rounds hurt (Section 5.6)

Results

Arg-C score, large PLM setting



Strengths

- Models how roles relate. PAIE predicts each role alone.
- Links the general pattern to the document. The graphs in Li et al. (2023) don't.
- Builds role vectors from each document. HMPEAE uses fixed prototypes.
- Careful ablations: edge types, update methods, rounds, role fusion

Weaknesses

- Small gains. On RAMS Arg-I, EPIG-EAE (58.1) is below HMPEAE (58.6) and about even with DEEIA (58.0).
- More rounds hurt after round 2. TabEAE decodes in one pass.
- Weak on OEE-CFC-style data (one event type, many roles). The authors say the pattern graph adds noise.
- No cost comparison with simpler models like PAIE.
- One trigger at a time. TabEAE models several events together.

Extensions

- Relation extraction: relation types as the pattern, entity pairs as the instance
- Dialogue slot filling: an intent's slots as the pattern, spans across turns as the instance
- Financial and legal clauses: the authors already have a financial dataset
- Cross-document event coreference: use each document's role vectors in a bigger graph

Takeaways

- EPIG links the general event pattern to the document
- Uses graph attention, role memory, and multiple rounds
- Good results, small gains, open questions on cost and scalability

Questions?